



Figure 2: Project structure



Figure 3: Hello World



Figure 4: Accelerometer

1. As you make your PhoneGap apps in HTML, CSS and JS, create an HTML page saying, 'Hello World', place it in its appropriate place and run it on your device. Let's create a very simple HTML page first:

```
// index.html
<!DOCTYPE html>
<html>
<head>
  <title>My Page</title>
  <meta name="viewport" content="width=device-width,
initial-scale=1">
  <script src="cordova-2.7.0.js"></script>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

So in this HTML page, there is a `<head>` section in which you have embedded your *phonegap.js* file and in the `<body>` section there is a heading tag with 'Hello World', which is going to be displayed when you run this app on the device.

2. Now, let's modify *MyPhoneGapActivity.java* to provide it the path of the HTML page you just created. The few changes that need to be made are:
 - Add the package name `'import com.phonegap.DroidGap';`
 - Modify the class statement and let it extend *DroidGap* instead of *Activity* – `'public class MyPhoneGapActivity extends DroidGap'`
 - Replace `setContentView()` with `super.loadUrl("file:///android_asset/www/index.html");`

The rest will be the same.
3. Now let's get ready to do some configuration in *AndroidManifest.xml* and add some permissions.

- Add the following code as a direct child of root `<manifest>`. This will give your application the flexibility to attain different screen sizes for different devices.

```
<supports-screens
  android:largeScreens="true"
  android:normalScreens="true"
  android:smallScreens="true"
  android:resizeable="true"
  android:anyDensity="true" />
```

- Now to enable your application to access some of the device features like the camera, GPS, Internet, etc, some permissions need to be given. This can be done by adding *uses-permission* either by using the *Permissions* tab that you can see on opening *AndroidManifest.xml* or by adding the following code as a child of `<manifest>`

```
<uses-permission android:name="android.
permission.CAMERA" />
<uses-permission android:name="android.permission.
VIBRATE" />
<uses-permission android:name="android.permission.
ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.
ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.
ACCESS_LOCATION_EXTRA_COMMANDS" />
<uses-permission android:name="android.permission.
READ_PHONE_STATE" />
<uses-permission android:name="android.permission.
INTERNET" />
<uses-permission android:name="android.permission.
RECEIVE_SMS" />
<uses-permission android:name="android.permission.
RECORD_AUDIO" />
<uses-permission android:name="android.permission.
MODIFY_AUDIO_SETTINGS" />
<uses-permission android:name="android.permission.
READ_CONTACTS" />
<uses-permission android:name="android.permission.
WRITE_CONTACTS" />
<uses-permission android:name="android.permission.
WRITE_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.
ACCESS_NETWORK_STATE" />
<uses-permission android:name="android.permission.
GET_ACCOUNTS" />
<uses-permission android:name="android.permission.
BROADCAST_STICKY" />
```

Now you're all set to deploy and run the application in the emulator/device.

4. To deploy and run, start the emulator, go to *Windows>Avd*